chapter

22 Biosynthesis of Amino Acids, Nucleotides, and Related Molecules

1. **ATP Consumption by Root Nodules in Legumes** Bacteria residing in the root nodules of the pea plant consume more than 20% of the ATP produced by the plant. Suggest why these bacteria consume so much ATP.

   Answer Bacteria in the root nodules maintain a symbiotic relationship with the plant: the plant supplies ATP and reducing power, and the bacteria supply ammonium ion by reducing atmospheric nitrogen. This reduction requires large quantities of ATP.

2. **Glutamate Dehydrogenase and Protein Synthesis** The bacterium *Methylophilus methylotrophus* can synthesize protein from methanol and ammonia. Recombinant DNA techniques have improved the yield of protein by introducing into *M. methylotrophus* the glutamate dehydrogenase gene from *E. coli.* Why does this genetic manipulation increase the protein yield?

   Answer The synthesis of protein requires the synthesis of amino acids. The transfer of nitrogen from an ammonium ion to carbon skeletons—that is, amino acid synthesis—can be carried out in two ways: (1) combination of the NH_3 with glutamate to form glutamine, catalyzed by glutamine synthetase and (2) reductive amination of α-ketoglutarate to form glutamate, catalyzed by glutamate dehydrogenase. The latter process, which is promoted by the introduction of the *E. coli* enzyme, is especially important because glutamate is the amino group donor in all transamination reactions.

3. **PLP Reaction Mechanisms** Pyridoxal phosphate can help catalyze transformations one or two carbons removed from the α carbon of an amino acid. The enzyme threonine synthase (see Fig. 22–15) promotes the PLP-dependent conversion of phosphohomoserine to threonine. Suggest a mechanism for this reaction.

   Answer A link between enzyme-bound PLP and the phosphohomoserine substrate is first formed, with rearrangement to generate the ketimine at the α carbon of the substrate. This activates the β carbon for proton abstraction, leading to displacement of the phosphate and formation of a double bond between the β and γ carbons. A rearrangement (beginning with proton abstraction at the pyridoxal carbon adjacent to the substrate amino nitrogen) moves the double bond between the α and β carbons, and converts the ketimine to the aldimine form of PLP. Attack of water at the β carbon is then facilitated by the linked pyridoxal, followed by hydrolysis of the imine link between PLP and the product, to generate threonine.

4. **Transformation of Aspartate to Asparagine** There are two routes for transforming aspartate to asparagine at the expense of ATP. Many bacteria have an asparagine synthetase that uses ammonium ion as the nitrogen donor. Mammals have an asparagine synthetase that uses glutamine as the nitrogen donor. Given that the latter requires an extra ATP (for the synthesis of glutamine), why do mammals use this route?

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Answer Recall that the ammonium ion is highly toxic to higher animals, especially to brain tissue. As NH_4^+ ions are transformed to glutamine, circulating NH_4^+ levels are reduced and toxic levels avoided.

5. Equation for the Synthesis of Aspartate from Glucose Write the net equation for the synthesis of aspartate (a nonessential amino acid) from glucose, carbon dioxide, and ammonia.

Answer We can approach this problem by working "backward" from aspartate to glucose as follows. Aspartate is synthesized from oxaloacetate by transamination from glutamate; glutamate is synthesized from α-ketoglutarate by glutamate dehydrogenase:

$$\text{Oxaloacetate} + \text{glutamate} \longrightarrow \text{aspartate} + \alpha\text{-ketoglutarate}$$
$$\alpha\text{-Ketoglutarate} + NH_3 + 2H^+ + \text{NADH} \longrightarrow \text{glutamate} + NAD^+ + H_2O$$

The sum of these reactions is

$$\text{Oxaloacetate} + NH_3 + 2H^+ + \text{NADH} \longrightarrow \text{aspartate} + NAD^+ + H_2O$$

Recall from Chapter 16 that oxaloacetate is synthesized from pyruvate by pyruvate carboxylase, and from Chapter 14 that pyruvate is produced from glucose via glycolysis:

$$\text{Pyruvate} + CO_2 + \text{ATP} + H_2O \longrightarrow \text{oxaloacetate} + \text{ADP} + P_i + 2H^+$$
$$\text{Glucose} + 2NAD^+ + 2\text{ADP} + 2P_i \longrightarrow 2 \text{ pyruvate} + 2\text{NADH} + 2H^+ + 2\text{ATP} + 2H_2O$$

Thus, we can write the net equation for aspartate synthesis:

$$\text{Glucose} + 2CO_2 + 2NH_3 \longrightarrow 2 \text{ aspartate} + 2H^+ + 2H_2O$$



6. Asparagine Synthetase Inhibitors in Leukemia Therapy Mammalian asparagine synthetase is a glutamine-dependent amidotransferase. Efforts to identify an effective inhibitor of human asparagine synthetase for use in chemotherapy for patients with leukemia has focused not on the amino-terminal glutaminase domain but on the carboxyl-terminal synthetase active site. Explain why the glutaminase domain is not a promising target for a useful drug.

Answer The amino-terminal glutaminase domain is quite similar in *all* glutamine amidotransferases. A drug that targeted this active site would probably inhibit many enzymes and thus be prone to producing many more side effects than a more specific inhibitor targeting the unique carboxyl-terminal synthetase active site.

7. Phenylalanine Hydroxylase Deficiency and Diet Tyrosine is normally a nonessential amino acid, but individuals with a genetic defect in phenylalanine hydroxylase require tyrosine in their diet for normal growth. Explain.

Answer In animals, tyrosine is synthesized from phenylalanine by phenylalanine hydroxylase. If this enzyme is defective, the biosynthetic route to tyrosine is blocked and this amino acid must be obtained from the diet.

8. Cofactors for One-Carbon Transfer Reactions Most one-carbon transfers are promoted by one of three cofactors: biotin, tetrahydrofolate, or *S*-adenosylmethionine (Chapter 18). *S*-Adenosylmethionine is generally used as a methyl group donor; the transfer potential of the methyl group in N^5-methyltetrahydrofolate is insufficient for most biosynthetic reactions. However, one example of the use of N^5-methyltetrahydrofolate in methyl group transfer is in methionine formation by the methionine synthase reaction (step ⑨ of Fig. 22–15); methionine is the immediate precursor of *S*-adenosylmethionine (see Fig. 18–18). Explain how the methyl group of *S*-adenosylmethionine can be derived from N^5-methyltetrahydrofolate, even though the transfer potential of the methyl group in N^5-methyltetrahydrofolate is one one-thousandth of that in *S*-adenosylmethionine.

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Answer The transfer potential of the methyl group of N^5-methyltetrahydrofolate is quite sufficient for the synthesis of methionine, which has an even lower methyl group transfer potential. The methyl group of methionine is activated by addition of the adenosyl group from ATP, converting methionine to *S*-adenosylmethionine (see Fig. 18–18). Recall that adoMet synthesis is one of only two known biochemical reactions in which triphosphate is released from ATP. Hydrolysis of the triphosphate renders the reaction thermodynamically more favorable.

9. **Concerted Regulation in Amino Acid Biosynthesis** The glutamine synthetase of *E. coli* is independently modulated by various products of glutamine metabolism (see Fig. 22–6). In this concerted inhibition, the extent of enzyme inhibition is greater than the sum of the separate inhibitions caused by each product. For *E. coli* grown in a medium rich in histidine, what would be the advantage of concerted inhibition?

Answer Because the regulatory mechanism is concerted, the amount of inhibition caused by saturating concentrations of histidine is limited—that is, a large excess of one amino acid does not shut down the flow of glutamate to glutamine. Metabolite flow continues, albeit at a reduced rate. If the inhibition of glutamine synthase were *not* concerted, saturating concentrations of histidine would shut down the enzyme and cut off production of glutamine, which the bacterium needs to synthesize other products.



10. **Relationship between Folic Acid Deficiency and Anemia** Folic acid deficiency, believed to be the most common vitamin deficiency, causes a type of anemia in which hemoglobin synthesis is impaired and erythrocytes do not mature properly. What is the metabolic relationship between hemoglobin synthesis and folic acid deficiency?

Answer Folic acid is a precursor of the coenzyme tetrahydrofolate (Fig. 18–16), which is required in the biosynthesis of glycine (Fig. 22–12). Because glycine is a precursor of porphyrins, the heme component of hemoglobin, a folic acid deficiency results in an impairment of hemoglobin synthesis, especially if the diet is also low in glycine.

11. **Nucleotide Biosynthesis in Amino Acid Auxotrophic Bacteria** Wild-type *E. coli* cells can synthesize all 20 common amino acids, but some mutants, called amino acid auxotrophs, are unable to synthesize a specific amino acid and require its addition to the culture medium for optimal growth. Besides their role in protein synthesis, some amino acids are also precursors for other nitrogenous cell products. Consider the three amino acid auxotrophs that are unable to synthesize glycine, glutamine, and aspartate, respectively. For each mutant, what nitrogenous products other than proteins would the cell fail to synthesize?

Answer Glycine, glutamine, and aspartate are required for the de novo synthesis of purine nucleotides; aspartate for the de novo synthesis of UMP; aspartate and glutamine for the de novo synthesis of CTP. Thus, glycine auxotrophs would fail to synthesize adenine and guanine nucleotides. Glutamine auxotrophs would fail to synthesize adenine, guanine, and cytosine nucleotides. Aspartate auxotrophs would fail to synthesize adenine, guanine, cytosine, and uridine nucleotides.

12. **Inhibitors of Nucleotide Biosynthesis** Suggest mechanisms for the inhibition of **(a)** alanine racemase by L-fluoroalanine and **(b)** glutamine amidotransferases by azaserine.

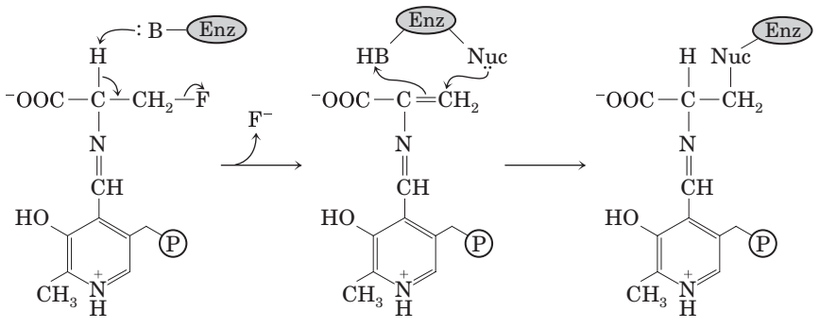
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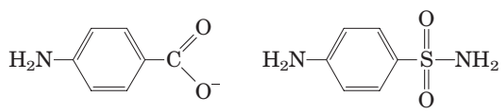
(a) See Figure 18–6, step ②, for the reaction mechanism of amino acid racemization. The F atom of fluoroalanine is an excellent leaving group. Fluoroalanine causes irreversible (covalent) inhibition of alanine racemase. One plausible mechanism is:



Nuc denotes any nucleophilic amino acid side chain in the enzyme active site.

(b) Azaserine (see Fig. 22–48) is an analog of glutamine. The diazoacetyl group is highly reactive and forms covalent bonds with nucleophiles at the active site of glutamine amidotransferases.

13. Mode of Action of Sulfa Drugs Some bacteria require *p*-aminobenzoate in the culture medium for normal growth, and their growth is severely inhibited by the addition of sulfanilamide, one of the earliest sulfa drugs. Moreover, in the presence of this drug, 5-aminoimidazole-4-carboxamide ribonucleotide (AICAR; see Fig. 22–33) accumulates in the culture medium. These effects are reversed by addition of excess *p*-aminobenzoate.



p-Aminobenzoate Sulfanilamide

(a) What is the role of *p*-aminobenzoate in these bacteria? (Hint: see Fig. 18–16.)

(b) Why does AICAR accumulate in the presence of sulfanilamide?

(c) Why are the inhibition and accumulation reversed by addition of excess *p*-aminobenzoate?

Answer

(a) *p*-Aminobenzoate is a component of tetrahydrofolate (see Fig. 18–16) and its derivative, N^5,N^{10}-methylenetetrahydrofolate, the cofactor involved in the transfer of one-carbon units.

(b) Sulfanilamide is a structural analog of *p*-aminobenzoate. In the presence of sulfanilamide, bacteria are unable to synthesize tetrahydrofolate, a cofactor necessary for the transformation of AICAR to *N*-formylaminoimidazole-4-carboxamide ribonucleotide (FAICAR) by the addition of —CHO; thus, AICAR accumulates.

(c) Excess *p*-aminobenzoate reverses the growth inhibition and ribonucleotide accumulation by competing with sulfanilamide for the active site of the enzyme involved in tetrahydrofolate biosynthesis. The competitive inhibition by sulfanilamide is overcome by the addition of excess substrate (*p*-aminobenzoate).

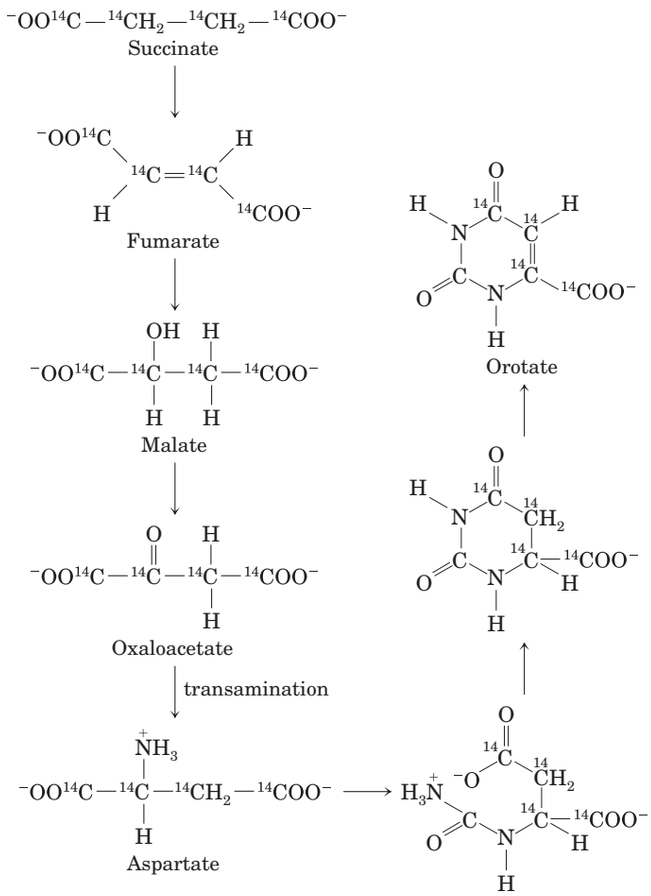
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14. Pathway of Carbon in Pyrimidine Biosynthesis Predict the locations of ^{14}C in orotate isolated from cells grown on a small amount of uniformly labeled [^{14}C]succinate. Justify your prediction.

Answer



15. Nucleotides as Poor Sources of Energy Under starvation conditions, organisms can use proteins and amino acids as sources of energy. Deamination of amino acids produces carbon skeletons that can enter the glycolytic pathway and the citric acid cycle to produce energy in the form of ATP. Nucleotides, on the other hand, are not similarly degraded for use as energy-yielding fuels. What observations about cellular physiology support this statement? What aspect of the structure of nucleotides makes them a relatively poor source of energy?

Answer Organisms do not store nucleotides to be used as fuel, and they do not completely degrade them, but rather hydrolyze them to release the bases, which can be recovered in salvage pathways. The low C:N ratio of nucleotides makes them poor sources of energy.

16. Treatment of Gout Allopurinol (see Fig. 22–47), an inhibitor of xanthine oxidase, is used to treat chronic gout. Explain the biochemical basis for this treatment. Patients treated with allopurinol sometimes develop xanthine stones in the kidneys, although the incidence of kidney damage is much lower than in untreated gout. Explain this observation in the light of the following solubilities in urine: uric acid, 0.15 g/L; xanthine, 0.05 g/L; and hypoxanthine, 1.4 g/L.

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Answer Treatment with allopurinol has two biochemical consequences. (1) It inhibits conversion of hypoxanthine to uric acid, causing accumulation of hypoxanthine, which is more soluble than uric acid and more readily excreted. This alleviates the clinical problems associated with AMP degradation. (2) It inhibits conversion of guanine to uric acid, causing accumulation of xanthine, which is even less soluble than uric acid. This is the source of xanthine stones. Because less GMP than AMP is degraded, kidney damage caused by xanthine stones is less than that caused by untreated gout.

17. Inhibition of Nucleotide Synthesis by Azaserine The diazo compound *O*-(2-diazoacetyl)-L-serine, known also as azaserine (see Fig. 22–48), is a powerful inhibitor of glutamine amidotransferases. If growing cells are treated with azaserine, what intermediates of nucleotide biosynthesis will accumulate? Explain.

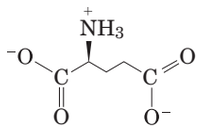
Answer In the de novo pathway of purine biosynthesis, the first step that requires glutamine is the conversion of 5-phosphoribosyl-1-pyrophosphate (PRPP) to 5-phospho-β-D-ribosylamine. In the presence of azaserine, which inhibits this conversion, PRPP accumulates.

Data Analysis Problem

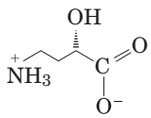
18. Use of Modern Molecular Techniques to Determine the Synthetic Pathway of a Novel Amino Acid Most of the biosynthetic pathways described in this chapter were determined before the development of recombinant DNA technology and genomics, so the techniques were quite different from those that researchers would use today. Here we explore an example of the use of modern molecular techniques to investigate the pathway of synthesis of a novel amino acid, (2*S*)-4-amino-2-hydroxybutyrate (AHBA). The techniques mentioned here are described in various places in the book; this problem is designed to show how they can be integrated in a comprehensive study.

AHBA is a γ-amino acid that is a component of some aminoglycoside antibiotics, including the antibiotic butirosin. Antibiotics modified by the addition of an AHBA residue are often more resistant to inactivation by bacterial antibiotic-resistance enzymes. As a result, understanding how AHBA is synthesized and added to antibiotics is useful in the design of pharmaceuticals.

In an article published in 2005, Li and coworkers describe how they determined the synthetic pathway of AHBA from glutamate.



Glutamate



AHBA

(a) Briefly describe the chemical transformations needed to convert glutamate to AHBA. At this point, don't be concerned about the *order* of the reactions.

Li and colleagues began by cloning the butirosin biosynthetic gene cluster from the bacterium *Bacillus circulans*, which makes large quantities of butirosin. They identified five genes that are essential for the pathway: *btrI*, *btrJ*, *btrK*, *btrO*, and *btrV*. They cloned these genes into *E. coli* plasmids that allow overexpression of the genes, producing proteins with "histidine tags" (see p. 314) fused to their amino termini to facilitate purification.

The predicted amino acid sequence of the BtrI protein showed strong homology to known acyl carrier proteins (see Fig. 21–5). Using mass spectrometry (see Box 3–2), Li and colleagues found a molecular mass of 11,812 for the purified BtrI protein (including the His tag). When the purified BtrI was incubated with coenzyme A and an enzyme known to attach CoA to other acyl carrier proteins, the majority molecular species had an M_r of 12,153.

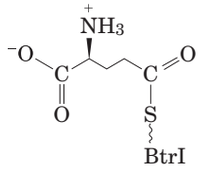
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(b) How would you use these data to argue that BtrI can function as an acyl carrier protein with a CoA prosthetic group?

Using standard terminology, Li and coauthors called the form of the protein lacking CoA apo-BtrI and the form with CoA (linked as in Fig. 21–5) holo-BtrI. When holo-BtrI was incubated with glutamine, ATP, and purified BtrJ protein, the holo-BtrI species of M_r 12,153 was replaced with a species of M_r 12,281, corresponding to the thioester of glutamate and holo-BtrI. Based on these data, the authors proposed the following structure for the M_r 12,281 species (γ-glutamyl-*S*-BtrI):



γ-Glutamyl-*S*-BtrI

(c) What other structure(s) is (are) consistent with the data above?

(d) Li and coauthors argued that the structure shown here (γ-glutamyl-*S*-BtrI) is likely to be correct because the α-carboxyl group must be removed at some point in the synthetic process. Explain the chemical basis of this argument. (Hint: See Fig. 18–6c.)

The BtrK protein showed significant homology to PLP-dependent amino acid decarboxylases, and BtrK isolated from *E. coli* was found to contain tightly bound PLP. When γ-glutamyl-*S*-BtrI was incubated with purified BtrK, a molecular species of M_r 12,240 was produced.

(e) What is the most likely structure of this species?

(f) Interestingly, when the investigators incubated glutamate and ATP with purified BtrI, BtrJ, and BtrK, they found a molecular species of M_r 12,370. What is the most likely structure of this species? Hint: Remember that BtrJ can use ATP to γ-glutamylate nucleophilic groups.

Li and colleagues found that BtrO is homologous to monooxygenase enzymes (see Box 21–1) that hydroxylate alkanes, using FMN as a cofactor, and BtrV is homologous to an NAD(P)H oxidoreductase. Two other genes in the cluster, *btrG* and *btrH*, probably encode enzymes that remove the γ-glutamyl group and attach AHBA to the target antibiotic molecule.

(g) Based on these data, propose a plausible pathway for the synthesis of AHBA and its addition to the target antibiotic. Include the enzymes that catalyze each step and any other substrates or cofactors needed (ATP, NAD, etc.).

Answer

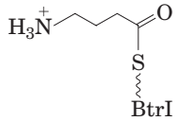
(a) The α-carboxyl group is removed and an —OH is added to the γ carbon.

(b) BtrI has sequence homology with acyl carrier proteins. The molecular weight of BtrI increases when incubated under conditions in which CoA could be added to the protein. Adding CoA to a Ser residue would replace an —OH (formula weight (FW) 17) with a 4′-phosphopantetheine group (see Fig. 21–5, p. 809). This group has the formula $C_{11}H_{21}N_2O_7PS$ (FW 356). Thus, 11,182 − 17 + 356 = 12,151, which is very close to the observed M_r of 12,153.

(c) The thioester could form with the α-carboxyl group.

(d) In the most common reaction for removing the α-carboxyl group of an amino acid (see Fig. 18–6, Ⓒ, p. 679), the carboxyl group must be free. Furthermore, it is difficult to imagine a decarboxylation reaction starting with a carboxyl group in its thioester form.

(e) 12,240 − 12,281 = 41, close to the M_r of CO_2 (44). Given that BtrK is probably a decarboxylase, its most likely structure is the decarboxylated form:

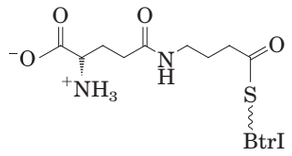


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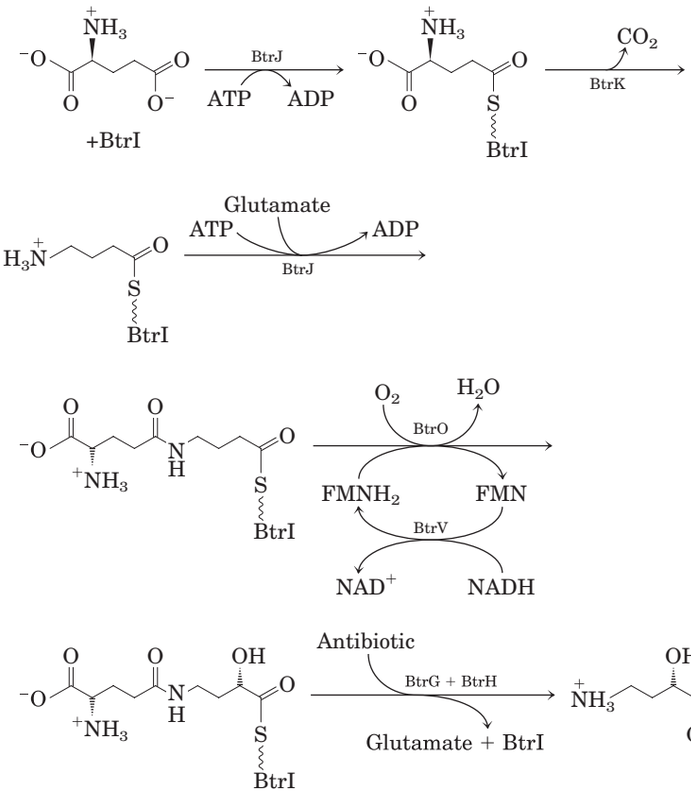
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(f) 12,370 − 12,240 = 130. Glutamic acid ($C_5H_9NO_4$; M_r 147), minus the —OH (FW 17) removed in the glutamylation reaction, leaves a glutamyl group of FW 130; thus, γ-glutamylating the molecule above would add 130 to its M_r. BtrJ is capable of γ-glutamylating other substrates, so it may γ-glutamylate the structure above. The most likely site for this is the free amino group, giving the following structure:



(g)



Reference

Li, Y., Llewellyn, N.M., Giri, R., Huang, F., & Spencer, J.B. (2005) Biosynthesis of the unique amino acid side chain of butirosin: possible protective-group chemistry in an acyl carrier protein–mediated pathway. *Chem. Biol.* **12,** 665–675.

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